

PARSA BASHARI

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Education

Sharif University of Technology

B.Sc. in Computer Engineering

- **GPA: 19.31** out of 20 (**Major: 19.58**)

Sep. 2021 – Present

Tehran, Iran

Research Interests

- Computer Architecture
- Domain-Specific Accelerators
- Processing-in-Memory
- Machine Learning

Research Experience

Barcelona Supercomputing Center

Junior Researcher

Aug. 2025 – Present

Barcelona, Spain

- *Computer Architecture for Parallel Paradigms (CAPP)* group, Supervisors: Prof. Osman Unsal & Prof. Adrian Cristal.
- Working on the implementation of a dedicated hardware unit for efficient execution of our proposed tiling algorithm for sparse tensor algebra acceleration.

Sharif University of Technology

Research Assistant

Jul. 2023 – Jun. 2025

Tehran, Iran

- *High-Performance Computing Architecture and Network (HPCAN)* group, Supervisor: Prof. Hamid Sarbazi-Azad.
- Worked on tiling in sparse tensor algebra acceleration (collaborating with Dr. Hajar Falahati & Dr. Negin Mahani).
- Contributed to two other projects: (1) *CSA-PIM*: Exploiting carry-save adder to reduce shift operations in Processing-in-Memory systems based on DRAM (collaborating with Dr. Nezam Rohbani); and (2) *Fullex*: Improving the representation of sparse tensors to reduce the number of intermediate matrices in outer-product-based matrix multiplication.

Publications

- **Parsa Bashari**, Hajar Falahati, Negin Mahani, Adrian Cristal, Osman Unsal, and Hamid Sarbazi-Azad. *Smartile: A Dynamic Sparsity-Pattern-Aware Tiling Strategy for Sparse Tensor Algebra Acceleration*. Under review (ISCA 2026).
- Danial Farsi, Mahdi Pazooki, **Parsa Bashari**, Hajar Falahati, Negin Mahani, Hakem Beitolahi, Adrian Cristal, and Osman Unsal. *Fullex: A Full-Dense Approach to Matrix Multiplication in Sparse Structures*. [SPICE 2025 Workshop](#), co-located with MICRO 2025, Seoul, Korea, October 18, 2025 (peer-reviewed workshop paper).

Teaching Experience

Sharif University of Technology

Teaching Assistant

Sep. 2022 – Jun. 2025

Tehran, Iran

- **Computer Architecture** — Dr. Laleh Arshadi Spring 2024 & Fall 2024 & Spring 2025
- **Machine Learning** — Dr. Ali Sharifi-Zarchi Fall 2024
- **Computer Networks** — Dr. Kambiz Mizanian Fall 2024
- **Artificial Intelligence** — Dr. Mohammad Hossein Rohban Spring 2024 & Fall 2024
- **Theory of Formal Languages and Automata** — Dr. Mohammad Izadi Spring 2024 & Spring 2025
- **Logic Design** — Dr. Shaahin Hessabi & Dr. Laleh Arshadi Spring 2024 & Spring 2025
- **Linear Algebra** — Dr. Maryam Ramezani Fall 2023 & Spring 2024
- **Computer Structure and Machine Language** — Prof. Hamid Sarbazi-Azad Fall 2023 & Spring 2025
- **Data Structures and Algorithms** — Dr. Mohammad Ali Abam Fall 2023
- **Engineering Probability and Statistics** — Dr. Amir Najafi Fall 2023
- **Advanced Programming in Java** — Dr. Mohammad Amin Fazli Spring 2023
- **Fundamentals of Programming in Python** — Dr. Iman Khani Jazani Spring 2023
- **Fundamentals of Programming in C** — Dr. Mohammad Amin Fazli Fall 2022

Honors and Awards

Ranked 28th out of 126,740 students (top 0.02%), Nationwide University Entrance Exam (Konkur) in Math

2021

Projects

- Smartile** | *Python, Verilog, L^AT_EX* |  **GitHub** (private for double-blind review process) **Summer 2024 – Present**
- Code of our paper *Smartile: A Dynamic Sparsity-Pattern-Aware Tiling Strategy for Sparse Tensor Algebra Acceleration*.
 - A dedicated hardware unit for efficient execution of our proposed tiling algorithm, implemented in Verilog, tested with ModelSim, and synthesized using Synopsys Design Compiler.
 - A sparse tensor algebra accelerator simulated by a model developed in Python, with the purpose of evaluating the effect of different tiling strategies on metrics such as execution time, buffer utilization, data reuse, and tile occupancies.
- DMA Module** | *Verilog* |  **GitHub** **Fall 2024**
- Homework assignment of *Interface Circuits Design* course.
 - An implementation of a basic system with direct memory access (DMA) module in Verilog (memory-mapped I/O, cycle-stealing approach for DMA module accessing the bus).
- Simple PIM System** | *Verilog* |  **GitHub** **Fall 2024**
- Project of *Modern VLSI Design* course.
 - A near-memory processing system implemented in Verilog, with support of addition, subtraction, bitwise OR, and bitwise AND operations. Full documentation available on GitHub.
- Maximal Independent Set** | *CUDA* |  **GitHub** **Fall 2023**
- Project of *Multi-core Computing* course.
 - The MIS problem is totally serial in nature and is not easy to parallelize, but our CUDA implementation reaches 250× speedup over the serial version for a graph with 1 million vertices. The full documentation is available on GitHub.
- Bellman-Ford and Matrix Multiplication** | *CUDA* |  **GitHub** **Fall 2023**
- CUDA implementation of Bellman-Ford algorithm and matrix multiplication with three approaches: inner product, outer product, and fine-grained parallelism with atomic accumulation.
- Linux Tutorial** | *Markdown* |  **GitHub** ☆ 40 **Summer 2023**
- A Linux tutorial/cheat sheet in markdown format.
- MIPS Processor** | *Logisim-evolution* |  **GitHub** **Spring 2023**
- Project of *Computer Architecture* course. The best project among 17 teams in the class.
 - **Specifications:** 2-way set-associative cache, 5-stage pipeline, data forwarding, branch prediction (saturating counter).
- Computer Structure and Machine Language Course Project** | *IBM360, 8086, MIPS* |  **GitHub** **Fall 2022**
- 10 assembly problems implemented in IBM360, 8086, and MIPS.
- Sid Meier's Civilization VI** | *Java, JavaFX, CSS* |  **GitHub** **Spring 2022**
- Project of *Advanced Programming* course.
- State.io** | *C, SDL2* |  **GitHub** **Fall 2021**
- Project of *Fundamentals of Programming* course.

Coursework

System-on-Chip Design (Graduate Level)	Dr. Shaahin Hessabi	18.5/20.0
Low Power Design (Graduate Level)	Dr. Alireza Ejlali	Grades Pending
Computer Architecture	Prof. Hamid Sarbazi-Azad	20.0/20.0
Multi-core Computing	Dr. Hajar Falahati	20.0/20.0
Modern VLSI Design	Dr. Shaahin Hessabi	19.5/20.0
Interface Circuits	Dr. Amin Foshati	20.0/20.0
Digital System Design	Dr. Alireza Ejlali	20.0/20.0
Computer Structure and Machine Language	Prof. Hamid Sarbazi-Azad	19.8/20.0
Machine Learning	Dr. Seyed Abolfazl Motahari	20.0/20.0
Artificial Intelligence	Dr. Mohammad Hossein Rohban	20.0/20.0
Quantum Computing (Graduate Level)	Dr. Mahdi Dolati	Grades Pending
Introduction to Bioinformatics	Dr. Ali Sharifi-Zarchi	Audited
Signals and Systems	Dr. Hossein Sameti	18.2/20.0
Linear Algebra	Dr. Maryam Ramezani	20.0/20.0
Probability and Statistics	Dr. Mahdi Jafari Siavoshani	20.0/20.0
Data Structures and Algorithms	Dr. Mahdi Safarnejad-Boroujeni	20.0/20.0

Technical Skills

General Programming Languages	C/C++, Python (Numpy, Pandas, Matplotlib, SciPy, Tensorflow, PyTorch), Java.
Special-purpose Languages	Verilog, CUDA, OpenMP, SQL.
Special-purpose Software Tools	ModelSim-Intel, Synopsys Design Compiler, Cadence SoC Encounter, Xilinx ISE.
Assembly Programming	x86, MIPS, IBM360.
Miscellaneous Tools	Linux, Git, L ^A T _E X, Markdown, Bash Scripting.

Academic Service

Byte Magazine | Website

Core Member

Sep. 2023 - Jun. 2024

Tehran, Iran

- The first official publication of Sharif University's CE department, which focuses on recent research and the edge of technology in computer science. I was a core member and a content writer in this publication since its inception.

ASM Masters | Website

Scientific Staff

Dec. 2023

Tehran, Iran

- ASM Masters is a competition based on assembly programming and machine code decoding.

HardWar | Website

Scientific Staff

Jun. 2023

Tehran, Iran

- HardWar is an annual competition held by Sharif University in a 2-day event, where students form teams and compete in different hardware-related topics.

Certificates and Online Courses

Machine Learning Specialization | Credential URL

Coursera

Stanford University

Issued Aug. 2023

- **Skills:** Supervised Learning, Linear Regression, Logistic Regression, Neural Networks, Decision Trees, Clustering, Anomaly Detection, Collaborative and Content-Based Filtering, PCA, Reinforcement Learning.
- I have shared my notes on this specialization (PDF and L^AT_EX code available on [Github](#)).

Mathematics for Machine Learning Specialization | Credential URL

Coursera

Imperial College London

Issued Sep. 2023

- These three courses were a brief practical refresh in my knowledge of the topics (I had passed multivariate calculus and linear algebra at a much deeper and more theoretical level in university).

Extracurricular Experience

Tehran's Rhetoric Orchestra

Violinist

Nov. 2024 – May 2025

Tehran, Iran

- Conductor: Reza Marivand, Position: second violins.

Iran Youth Symphonic Orchestra | YouTube

Violinist

Mar. 2024 – Mar. 2025

Tehran, Iran

- Conductor: Arman Noroozi, Position: first violins.

Music Center of Sharif University of Technology

Violin Tutor

Aug. 2022 – June 2024

Tehran, Iran

- I have been personally teaching classical violin to over 30 students.

Languages

English	Professional working proficiency - TOEFL iBT: 104 out of 120 (R: 30, L: 29, S: 21, W: 24)
Spanish	Elementary proficiency
Persian (Farsi)	Native proficiency